Chapter 1: Introduction to Strategic Cost Management

Question: 1

Which one of the following, simply answers the question "What else will accomplish the purpose of the product, service or process we are studying?"

(A) Value Engineering

(B) Value Analysis

(C) Business Process Re-engineering

(D) Supply Chain Strategy

(Dec 2025)

Question: 2

Which of the following is not a principle of Business Process Re-engineering?

(A) Organise work around outcomes, not tasks.

(B) Capture information once and at the source.

(C) Empower workers who use the process.

(D) Rely heavily on hierarchy and supervision.

(Dec 2025)

Question: 3

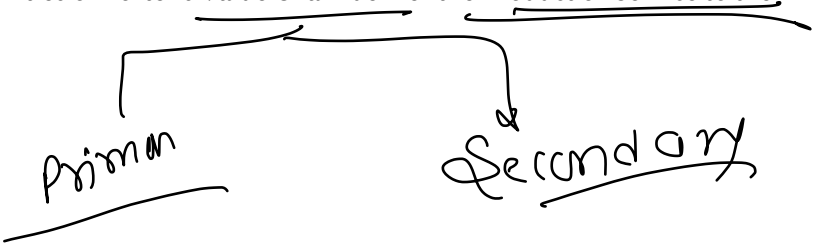
Which of the following primary activities of Porter's Value Chain deliver the Product or Service to the Customer?

(A) Operations

(B) Outbound Logistics

(C) Infrastructure

(D) Services



(Jun 2025)

Question: 4

Which of the following important pillars of Strategic Cost Management determines the company's comparative position in the industry in terms of performance?

(A) Cost Driver Analysis

(B) Value Chain Analysis

(C) Strategic Positioning Analysis

(D) Competitive Value Analysis

(Dec 2024)

Question: 5

Which one of the following is true for Blue Ocean Strategy?

(A) Beat the competition

(B) Exploit existing demand

(C) Make the competition irrelevant

(D) Make the value-cost trade off

(Jun 2024)

Question: 6

Which one of the following important pillars of Strategic Cost Management determines the company's comparative position in the Industry in terms of performance?

(A) Cost drivers Analysis

(B) Value Chain Analysis

(C) Strategic positioning analysis

(D) Competitive Value Analysis

(Dec 2023)

Question: 7

Which one of the following is not a support activity of the value chain?

(A) Human Resource Management

(B) Technological Development

(C) Service

(D) Infrastructure

(June 2023)

Question: 8

Which one of the following is not true for a Red ocean strategy and why?

(A) Beat the competition

(B) Exploit existing demand

(C) Make the value cost trade-off

(D) Break the value cost trade-off

(June 2023)

Question: 9

Activities required to design, develop, produce, market, distribute, and service a product are known as –

(A) target activities.

(B) value-chain activities.

(C) whole life activities.

(D) overhead.

(Dec 2021)

Question: 10

Stock Control data for Material P are:

Annual usage: 3600 units;

Cost per unit: ₹100/-;

Cost of placing an order: ₹40;

Stockholding Cost: 20% of the overall stock volume;

Lead time: One month

The EOQ based on the above data is:

(a) 210 units

(b) 175 units

$$EOQ = \sqrt{\frac{2AO}{C}}$$

$$= \sqrt{\frac{2 \times 3600 \times 40}{20\% \times 100}}$$

$$= 120$$

(c) 90 units
(d) 120 units

(June 2017)

1.	(A) Value Engineering
2.	(D) Rely heavily on hierarchy and supervision.
3.	(B) Outbound Logistics
4.	(C) Strategic Positioning Analysis
5.	(C) Make the competition irrelevant
6.	(C) Strategic positioning Analysis
7.	(C) Service
8.	(D) Break the value cost trade-off
9.	(B) value-chain activities.
10.	(D) 120 units **Explanation:** 120 units as per the following computation: EOQ = √2AB / C, where

Chapter 2: Quality Cost Management

Question: 1

Which of the following is/are not the quality parameter(s) for manufacturing organisations?

(A) Reliability

(B) Durability

(C) Service ability

(D) Courtesy/Friendliness

Dimensions of Quality for Manufacturing Versus Service Organizations

Manufacturing organizations	Service organizations
Conformance to specifications	Tangible factors
Performance	Consistency
Reliability	Responsiveness to customer needs
Features	Courtesy/friendliness
Durability	Timeliness/ promptness
Serviceability	Atmosphere

(Jun 2025)

Question: 2

Which of the following quality costs is incurred when inferior products are delivered to customers?

(A) Prevention Costs → making sure defectives won't arise

(B) Appraisal Costs → inspection

(C) External Failure Costs → Defective reached to customer

(D) Internal Failure Costs → Defective correction in-house

(Dec 2024)

Question: 3

Which of the following Quality Cost is incurred in the process of uncovering defects?

(A) Appraisal Costs

(B) Prevention Costs

(C) Internal Failure Costs

(D) External Failure Costs

(Jun 2024)

Question: 4

Down time due to defect in quality is an example of

(A) Internal failure cost

(B) Prevention cost

(C) Appraisal cost

(D) External failure cost

(Dec 2023)

Question: 5

Warranty period return of finished goods sold falls under the following quality cost:

(A) Prevention

(B) Appraisal

(C) Internal failure

(D) External failure

(June 2023)

Question: 6

Depreciation of product testing equipment is

(A) Internal Failure Cost

(B) Appraisal Cost

(C) Not a Quality Cost

(D) Irrelevant, but Quality Cost

(Dec 2022)

Question: 7

A factory is setting up a special inspection at the supply point of raw materials at Rs. 80,000. Consequent to this, there is lesser number of returns from customers. These goods used to be sold for Rs. 1,00,000 and variable costs are Rs. 80,000. The change in quality costs are –

(A) Decrease by Rs. 80,000

(B) Decrease by Rs. 60,000

(C) Decrease by Rs. 20,000

(D) No change

(Dec 2021)

Question: 8

Liability claims is an example of –

(A) prevention costs.

(B) appraisal costs.

(C) external failure costs.

(D) internal failure costs.

(Dec 2021)

Question: 9

X Ltd. has 1,000 units of an obsolete item which are carried in inventory at the original price of ₹ 50,000. If these items are reworked for ₹ 20,000, they can be sold for ₹ 36,000. Alternatively, they can be sold as scrap for ₹ 6,000 in the market. In a decision model used to analyse the reworking proposal, the opportunity cost should be taken as:

(A) ₹ 16,000

(B) ₹ 6,000

(C) ₹ 30,000

(D) ₹ 20,000

(June 2019)

Question: 10

Match the following:

(A)	Dr. Deming believes	(1) Common causes
(B)	Ishikawa development	(2) To prevent defect
(C)	Type of variation is due to	(3) Cause & effect diagram
(D)	Crosby's objective of quality	(4) Histogram

The correct order is

(a) A-3, B-2, C-1, D-4

– from manufacturing to transactional and from product to service.

Six Sigma has two key methodologies: DMAIC and DMADV, both inspired by W. Edwards Deming's Plan-Do-Check- Act Cycle: DMAIC is used to improve an existing business process, and DMADV is used to create new product or process designs for predictable, defect-free performance.

"Quality is free" - Phil Crosby. This is a unique theory when there are no defects, then there is no wastage and Boosted Sales, thus quality becomes free.

(b) A-2, B-3, C-4, D-1
(c) A-2, B-3, C-1, D-4
(d) A-4, B-3, C-1, D-2

(Dec 2017)

1.	(D) Courtesy/Friendliness
2.	(C) External Failure Costs
3.	(A) Appraisal Costs
4.	(A) Internal failure cost
5.	(D) External failure
6.	(B) Appraisal Cost
7.	(D) No change
8.	(C) external failure costs.
9.	(B) ₹ 6,000 Explanation: Original price is not relevant **Particulars** | **Amount (₹)** Rework income | 36,000 Less: Cost of rework | 20,000 **Net inflow** | **16,000 (it is relevant)** The other alternative relevant cash flow is from sale as scrap = **₹ 6,000** Hence the **opportunity cost is ₹ 6,000**
10.	(C) A-2, B-3, C-1, D-4

Chapter 3: Decision Making Techniques
(Case-Study Based Approach)

Question: 1

RIM Ltd. has estimated the demand level of its product:

Sales (units)	5,000	10,000	15,000	20,000
Probability	0.10	0.20	0.40	0.30

It has an estimated contribution of ₹ 4 per unit and total fixed costs of ₹ 60,000. What is the probability that the company will make a profit of at least ₹ 15,000?

(A) 40%

(B) 30%

(C) 20%

(D) None of the above

(Dec 2025)

Question: 2

SM Ltd., a manufacturing company produces a product which is sold at a price of ₹ 80 p.u. Its break even sales is ₹ 15 lakhs, while its fixed cost is ₹ 6 lakhs. What will be its desired quantity of sales to earn a profit of ₹ 4 lakhs?

(A) 18,750 units

(B) 31,250 units

(C) 32,320 units

(D) None of the above

(Dec 2025)

Question: 3

BOS Ltd., a manufacturing company, requires ₹ 90 lakhs in sales to meet target net profit of ₹ 15 lakhs. The company's contribution margin is 30%. What will be its fixed cost?

(A) ₹ 15 lakhs

(B) ₹ 12 lakhs

(C) ₹ 10 lakhs

(D) ₹ 20 lakhs

(Dec 2025)

Question: 4

The shadow price of skilled labour for SD Ltd. is currently ₹ 10 per hour. What does this mean?

(A) The cost of obtaining additional skilled labour is ₹ 10 per hour.

(B) There is a hidden cost of ₹ 10 for each hour of skilled labour actively worked.

(C) Contribution will be increased by ₹ 10 per hour for each extra hour of skilled labour that can be obtained.

(D) The total costs will be reduced by ₹ 10 for each additional hour of skilled labour that can be obtained.

(Dec 2025)

Question: 5

The product of XYZ company is sold at a fixed price of ₹ 1,500 per unit. As per company's estimate, 500 units of the product are expected to be sold in the coming year. If the value of investments of the company is ₹ 15 lakhs and it has a target ROI of 15%, the target cost would be

(A) ₹ 930

(B) ₹ 950

(C) ₹ 1,050

(D) ₹ 1,130

(Dec 2025)

Question: 6

The break-even point of a manufacturing company is ₹ 1,60,000. Fixed cost is ₹ 48,000.Variable cost is ₹ 12 per unit. The PV ratio will be

(A) 20%

(B) 40%

(C) 30%

(D) 25%

(Jun 2025)

Question: 7

A factory of MONZ Ltd., can make only one of the three products A, B or C in a given production period. The following informations are given:

Per Unit (₹)	A	B	C
Selling price	1,500	2,000	2,500
Variable Cost	700	1,100	1,500

Assume that there is no constraint on resource utilization or demand and similar resources are consumed by A, B and C. The opportunity cost of making one unit of C is

(A) ₹ 900

(B) ₹ 800

(C) ₹ 1,800

(D) ₹ 1,500

(Jun 2025)

Question: 8

SONT Ltd., has fixed costs of ₹ 5,00,000 and variable costs are 70% of the selling price. To realize profits of ₹ 1,00,000 from sales of 400000 units, the selling price per unit must be

(A) ₹ 5

(B) ₹ 6

(C) ₹ 7

(D) ₹ 4

(Jun 2025)

Question: 9

ROTIZ Ltd., a manufacturing company has a break-even point when sales are ₹ 12,00,000 and fixed costs at that level of sales are ₹ 4,80,000.

If the margin of safety (MOS) and sales of the company are 40% and 5,00,000 units respectively, what will be the sale price per unit?

(A) ₹ 6

(B) ₹ 5

(C) ₹ 4

(D) None of the above

(Dec 2024)

Question: 10

120 units of semi-conductors are required to be sold to earn a profit of ₹ 1,00,000 in a monopoly market. The fixed cost for the period is ₹ 80,000.The contribution in the monopoly market is as high as 3/4th of its variable cost. Determine the target selling price per unit.

(A) ₹ 4,500

(B) ₹ 3,250

(C) ₹ 4,000

(D) ₹ 3,500

(Dec 2024)

Question: 11

The Break Even Sales of GXT Ltd., a manufacturing company, is ₹ 3,75,000, while its fixed cost is ₹ 1,12,500. If the Margin of Safety is 40%, what will be its Profit?

(A) ₹ 80,000

(B) ₹ 75,000

(C) ₹ 64,000

(D) Insufficient information

(Jun 2024)

Question: 12

BENT Ltd., a manufacturer of components for VCD, has a capacity to produce 4 Lakh units. The market demand is sensitive to the sale price and the company could sell 1 Lakh units at a price of ₹ 5,000 each. The demand thereafter would double for each ₹ 500 per unit fall in the selling price. If the company expects a minimum margin of 25%, what would be the Target Cost per unit for the company to sell at full capacity utilization?

(A) ₹ 4,000

(B) ₹ 3,200

(C) ₹ 3,000

(D) ₹ 2,900

(Jun 2024)

Question: 13

A particular job MB requires 800 kgs. of a material. 500 kgs. of the particular material is currently in stock. The original price of the material was ₹ 300 but current resale value of the same has been determined as ₹ 200. The current replacement price of the material is Re. 0·80 per kg., what will be the relevant cost of material for job MB?

(A) ₹ 640

(B) ₹ 440

(C) ₹ 300

(D) None of the above

(Jun 2024)

Question: 14

The profit volume ratio of Divon Ltd. is 30%, while the margin of safety is 40%. If fixed cost of the company is ₹ 1,80,000, what will be its profit (net)?

(A) ₹ 1,20,000

(B) ₹ 1,50,000

(C) ₹ 1,60,000

(D) None of the above

(Dec 2023)

Question: 15

During which phase of the Product Life Cycle, sales rise at an increasing rate?

(A) Introduction

(B) Growth

(C) Maturity

(D) Decline

(Dec 2023)

Question: 16

An increase in the selling price per unit will cause

(A) a decrease in the number of units required to breakeven.

(B) an increase in the contribution margin ratio.

(C) an increase in the margin of safety.

(D) All of the above

(Dec 2023)

Question: 17

JACKSON Ltd., sells its product for ₹ 8,000 per unit. Variable costs per unit are – manufacturing ₹ 4,000 and selling and administrative ₹ 80. Fixed costs are ₹ 20,000 manufacturing overheads and ₹ 30,000 selling and administrative overheads. There was no beginning inventory. Production was 25 units during the month and sales was 20 units during the month. What will be the Income under Absorption Costing for the month?

(A) ₹ 32,400

(B) ₹ 30,800

(C) ₹ 30,000

(D) None of the above

(Dec 2023)

Question: 18

A production of ZON Ltd. has the capacity to produce either 4000 units of A, or 3500 units of B or 5000 units of C. Only one product can be made in a production period. The contributions per unit of A,B and C are ₹ 10, ₹ 11 and ₹ 8 respectively. The opportunity cost of A would be:

(A) ₹ 44,000
(B) ₹ 38,500
(C) ₹ 50,000
(D) ₹ 40,000

(Dec 2023)

Question: 19

ROBINSON Ltd., a manufacturing company has a break-even point, when sales are ₹ 10 lakh and fixed costs of ₹ 4 lakh. To realize profits of ₹ 2 lakh from sales of 300000 units, the selling price per unit will be:

(A) ₹ 6
(B) ₹ 5
(C) ₹ 4
(D) ₹ 2

(Dec 2023)

Question: 20

AMON Ltd., plans to introduce a new product ZOS and is using Target cost approach. The selling price of product ZOS is set at ₹ 120 for each unit and sales revenue for the coming year is expected to be ₹ 9,60,000. The Co. requires a return of 15% on the coming year on its investment of ₹ 20 lakh. What is the Target Cost per unit for the coming year?

(A) ₹ 90.00
(B) ₹ 85.00
(C) ₹ 82.50
(D) ₹ 80.50

(Dec 2023)

Question: 21

X LTD. has annual turnover of ₹ 200 lakhs and an average C/S ratio of 40%. It makes 10% profit on sales before charging depreciation and interest which amount to ₹ 10 lakhs and ₹ 15 lakhs respectively. The fixed cost of X LTD. will be

(A) ₹ 70 lakhs
(B) ₹ 85 lakhs
(C) ₹ 90 lakhs
(D) ₹ 60 lakhs

(June 2023)

Question: 22

TOTA LTD. has 200 units of an obsolete item which are carried in inventory at the original price of ₹ 36,000. If these items are reworked for ₹ 12,000, they can be sold for ₹ 21,600. Alternatively they can

be sold as scrap for ₹ 3,600 in the market. In a decision model used to analyse the reworking proposal, the opportunity cost should be taken as

(A) ₹ 4,000

(B) ₹ 3,600

(C) ₹ 3,000

(D) None of the above

(June 2023)

Question: 23

WYE LTD. is using the target cost approach. The following data has been collected:

Expected sales revenue: ₹ 75,00,000 (₹ 15 per unit)

Projected selling and distribution cost: ₹ 3,00,000

Projected Variable Cost per unit : ₹ 10

Desired Profit: ₹ 9,50,000

What will be the target cost per unit?

(A) ₹ 14.40

(B) ₹ 13.10

(C) ₹ 12.50

(D) None of the above

(June 2023)

Question: 24

ASH LTD. a manufacturing company produced 300 units of product-HB and sold 240 units @ ₹ 600 per unit during the year. The following information pertains to production and sale for the year:

Variable cost per unit	₹ 330
Fixed costs for the year:	
Manufacturing overheads	₹ 21,000
Selling and Administration Overheads	₹ 30,000

There was no beginning inventory.

What will be the income under absorption costing for the year?

(A) ₹ 24,000

(B) ₹ 18,000

(C) ₹ 15,000

(D) None of the above

(June 2023)

Question: 25

Target Costing is the answer to

(A) Market Driven Prices

(B) Seller's Market

(C) No Profit Situation

(D) None of the Above

(June 2023)

Question: 26

SINT Ltd. determine its selling price by marking up the variable cost 50%. In addition, the company uses frequent selling price mark down to stimulate sales. If the mark down average is 20%, what is the company's contribution margin Ratio?

(A) 16.67%

(B) 18.50%

(C) 20.30%

(D) None of the above

(June 2023)

Question: 27

The Income statement of BOMA Ltd. is summarized as below:

Particulars	₹ in Lakh
Net Revenue	120
Less: Expenses (including ₹ 60 lakh of fixed cost)	132
Net Loss	12

The Manager (F&A) believes that an increase of ₹ 30 lakh as fixed expenditure in advertising outlays will increase the sales substantially.

At what sales volume will the company have break even?

(A) ₹ 250 lakh

(B) ₹ 225 lakh

(C) ₹ 200 lakh

(D) None of the above

(Dec 2022)

Question: 28

SOVI Ltd., a mobile phone manufacturer, is planning to introduce a new mobile phone. The potential market over the next year is 10,00,000 units.

The SOVI Ltd. has the capacity to produce 4,00,000 units and could sell 1,00,000 units at ₹ 10,000 per unit.

Demand would double for each ₹ 1,000 fall in the selling price.

The company has a 25% profit margin on sales for the similar products.

What is SOVI Ltd.'s target cost per unit to the nearest Re?

(A) ₹ 8,000

(B) ₹ 7,000

(C) ₹ 6,000

(D) ₹ 4,000

(Dec 2022)

Question: 29

MN Ltd. sales for the years 2018 and 2019 were ₹ 8,00,000 and ₹ 7,92,000. The cost of goods sold correspondingly were ₹ 4,80,000 and ₹ 4,64,000. In the year 2019 selling price was reduced by 10%

as compared to 2018.

What was the decrease in gross profit caused by the change in selling price?

(A) ₹ 8,000

(B) ₹ 72,000

(C) ₹ 79,200

(D) ₹ 88,000

(Dec 2022)

Question: 30

The Holiday Card Company, a producer of specialty cards, has asked you to complete several calculations based upon the following information:

Particulars	Details
Income tax rate	30%
Selling price per unit (₹)	6.60
Variable cost per unit (₹)	5.28
Total fixed costs (₹)	46,200

How many cards must be sold to earn an after-tax net income of ₹ 18,480?

(A) 60,000 units

(B) 45,000 units

(C) 75,000 units

(D) 55,000 units

(Dec 2022)

Question: 31

In cost-plus pricing, the markup consists of

(A) total cost and desired ROI.

(B) selling and administrative costs.

(C) manufacturing costs.

(D) desired ROI.

(Dec 2021)

Question: 32

Only direct materials, direct labor, and variable manufacturing overhead costs are considered product costs when using –

(A) absorption costing.

(B) full costing.

(C) variable costing.

(D) product costing.

(Dec 2021)

Question: 33

When there is excess capacity, it makes sense to accept a one-time-only special order for less than the current selling price when –

(A) incremental revenues exceed incremental costs.

(B) additional fixed costs need not be incurred to accommodate the order.

(C) there is a positive contribution per unit of the product under normal capacity and spare capacity.

(D) the special order is from a normal customer.

(Dec 2021)

Question: 34

A company that is a price-taker would most likely use which of the following methods?

(A) Target costing

(B) Cost plus pricing, contribution approach

(C) Cost plus pricing, absorption approach

(D) Time-and-material pricing

(Dec 2021)

Question: 35

If the unit level of inventory increases during an accounting period, then –

(A) operating income will be the same under absorption costing and variable costing.

(B) the exact effect on operating income cannot be determined.

(C) more operating income will be reported under absorption costing than variable costing.

(D) less operating income will be reported under absorption costing than variable costing.

(Dec 2021)

Question: 36

_____ is the difference between the sales price needed to capture a predetermined market share and the desired profit per unit.

(A) Gross profit

(B) Target cost

(C) Target price

(D) None of these.

(Dec 2021)

Question: 37

The _____ is a period of time when sales increase at a decreasing rate.

(A) maturity stage

(B) growth stage

(C) introduction stage

(D) decline stage

(Dec 2021)

Question: 38

NM paid Rs. 5,30,000 for a machine used to powder wheat. The machine can be sold for Rs. 1,30,000. The sale value of wheat is Rs. 8,00,000 and its variable cost is Rs. 4,50,000. The opportunity cost of producing wheat flour is –

(A) Rs. 5,30,000

(B) Rs. 3,50,000
(C) Rs. 8,00,000
(D) Rs. 1,30,000

(Dec 2021)

Question: 39

Which of the following will always be a relevant cost?
(A) Fixed cost
(B) Opportunity cost
(C) Variable cost
(D) Sunk cost

(Dec 2021)

Question: 40

The break-even point of a manufacturing company is ₹1,60,000. Fixed cost is ₹48,000. Variable cost is ₹12 per unit. The PV ratio will be:
(A) 20%
(B) 40%
(C) 30%
(D) 25%

(Dec 2019)

Question: 41

What is the opportunity cost of making a component part in a factory given no alternative use of the capacity?
(A) The variable manufacturing cost of the component
(B) The total manufacturing cost of the component
(C) The total variable cost of the component
(D) Zero

Because nothing else could have been done with that time & space, nothing of value is given up. Therefore, opportunity cost is zero.

(Dec 2019)

Question: 42

The product of XYZ company is sold at a fixed price of ₹1,500 per unit.
As per company's estimate, 500 units of the product are expected to be sold in the coming year.
If the value of investments of the company is ₹15 lakh and it has a target ROI of 15%, the target cost would be:
(A) ₹930
(B) ₹950
(C) ₹1050
(D) ₹1130

1500
450
1050

sound

225000

(Dec 2019)

Question: 43

T Ltd. produces and sells a product. The company expects the following revenues and costs in 2018:

Particulars	Amount (₹)

Revenues (400 sets sold @ ₹600 per product)	2,40,000
Variable costs	1,60,000
Fixed costs	50,000

What amount of sales must T Ltd. have to earn a target net income of ₹ 63,000 if they have a tax rate of 30%?

(A) ₹ 4,20,000

(B) ₹ 4,29,000

(C) ₹ 3,00,000

(D) ₹ 4,89,000

(June 2019)

Question: 44

Excel Products Ltd. manufactures four products e.g. Product E, Product F, Product G and Product H using same raw materials. The input requirements for Products E, F, G and H are 1kg, 2kgs, 5kgs and 7kgs, respectively. Product-wise Selling Price and Variable Cost data are given hereunder:

Products	E	F	G	H
Selling Price (₹)	100	150	200	300
Variable Cost (₹)	50	70	100	125

Assuming raw material availability is a limiting factor, the correct ranking of the products would be:

(A) E, F, G & H

(B) E, F, H & G

(C) F, E, G & H

(D) F, E, H & G

(June 2019)

Question: 45

The shadow price of skilled labour for SD Ltd. is currently ₹10 per hour. What does this mean?

(A) The cost of obtaining additional skilled labour is ₹10 per hour.

(B) There is a hidden cost of ₹10 for each hour of skilled labour actively worked.

(C) Contribution will be increased by ₹10 per hour for each extra hour of skilled labour that can be obtained.

(D) The total costs will be reduced by ₹10 for each additional hour of skilled labour that can be obtained.

(June 2019)

Question: 46

A company has a break-even point when sales are Rs. 3,20,000 and variable cost at that level of sales are Rs. 2,00,000. How much would contribution margin increase or decrease if variable expenses are dropped by Rs. 30,000?

(A) Increase by 27.5%

(B) Increase by 9.375%

(C) Decrease by 9.375%

(D) Increase by 37.5%

(Dec 2018)

Question: 47

A company has the capacity of producing 80,000 units and presently sells 20,000 units at Rs. 100 each. The demand is sensitive to selling price and it has been observed that with every reduction of Rs. 10 in selling price the demand is doubled. What should be the target cost in selling price if the demand is doubled at full capacity and profit margin on sale is taken at 25%?

(A) Rs. 75

(B) Rs. 90

(C) Rs. 25

(D) Rs. 60

(Dec 2018)

Question: 48

A factory can make only one of the three products X, Y or Z in a given production period. The following information are given:

Per unit Rs.	X	Y	Z
Selling Price	1500	1800	2000
Variable Cost	700	950	1000

Assume that there is no constraint on resource utilization or demand and similar resources are consumed by X, Y and Z. The opportunity cost of making one unit of Z is

(A) Rs. 850

(B) Rs. 800

(C) Rs. 1800

(D) Rs. 1500

(Dec 2018)

Question: 49

AB company is a supermarket group that incurs the following costs:

(a) The bought-in price of the goods

(b) Inventory finance costs

(c) Self-refilling costs

(d) Costs of repacking or 'pack out' prior to storage before sale

AB company's calculating of direct product profit (DPP) would include

Costs (a) and (c), only.

All of the above cost except (b)

All of the above costs except (d)

All of the above costs.

(Dec 2018)

Question: 50

H Group has two divisions, Division P and Division Q. Division P manufactures an item that is transferred to Division Q. The item has no external market and 6000 units produced are transferred internally each year. The costs of each division are as follows:

	Division P	Division Q
Variable Cost	Rs. 100 per unit	Rs. 120 per unit
Fixed cost each year	Rs. 1,20,000	Rs. 90,000

Head Office management decided that a transfer price should be set that provides a profit of Rs. 30,000 to Division P. What should be the transfer price per unit?

(A) Rs. 145
(B) Rs. 125
(C) Rs. 120
(D) Rs. 135

(Dec 2018)

Question: 51

A Company requires ₹ 85,00,000 in sales to meet its target net profit. Its contribution margin is 30% and the fixed costs are ₹ 15,00,000. What is the target net profit?

(a) ₹ 10,50,000
(b) ₹ 19,50,000
(c) ₹ 25,50,000
(d) ₹ 35,00,000

(June 2018)

Question: 52

By making and selling 9,000 units of a product, a company makes a profit of ₹ 10,000, whereas in the case of 7,000 units, it would lose ₹ 10,000 instead. The number of units to break-even is

(a) 7,500 units
(b) 8,000 units
(c) 7,750 units
(d) 8,200 units

(June 2018)

Question: 53

1200 units of microchips are required to be sold to earn a profit of ₹ 1,06,000 in a monopoly market. The fixed cost for the period is ₹ 74,000. The contribution in the monopoly market is as high as 3/4th of its variable cost. Determine the target selling price per unit.

(a) 450
(b) 325
(c) 400
(d) 350

(June 2018)

Question: 54

If project A has a net present value (NPV) of ₹ 30,00,000 and project B has an NPV of ₹ 50,00,000, what is the opportunity cost if project B is selected?

(a) ₹ 23,00,000

(b) ₹ 30,00,000
(c) ₹ 20,00,000
(d) ₹ 50,00,000

(June 2018)

Question: 55

Empire Hotel has a capacity of 100 single rooms and 20 double rooms. Average occupancy is 70% for 365 days of the year. The rent for a double room is kept at 130% of a single room. The total room occupancy days in a year in terms of single room is

(a) 32193
(b) 30660
(c) 31660
(d) 30993

(June 2018)

Question: 56

A company produces a product which is sold at a price of ₹80. Its Variable cost is ₹32. The company's Fixed cost is ₹11,52,000 p.a. The company operates at a margin of safety of 40%. The total sales of the company is:

(a) 4,000 units
(b) 40,000 units
(c) 30,000 units
(d) 20,000 units

(Dec 2017)

Question: 57

The P/V ratio of a firm dealing in Electrical equipment is 50% and the margin of safety is 40%. BEP of the firm at a sales volume of ₹50,00,000 will be

(a) ₹25,00,000
(b) ₹35,00,000
(c) ₹30,00,000
(d) ₹36,00,000

(Dec 2017)

Question: 58

ABC Limited has current PBIT of ₹19.20 lakhs on total assets of ₹96 lakhs. The company has decided to increase assets by ₹24 lakhs, which is expected to increase the operating profit before depreciation by ₹8.40 lakhs. There will be a net increase in depreciation by ₹4.80 lakhs. This will result in ROI

(a) to increase by 1%
(b) to decrease by 1%
(c) to decrease by 1.5%
(d) to remain the same

(Dec 2017)

Question: 59

Marketing department of an organisation estimates that 40,000 of new mixers could be sold annually at a price of ₹60 each. To design, develop and produce these new mixers an investment of ₹40,00,000 would be required. The company desires a 15% return on investment (ROI). Given these data, the target cost to manufacture, sell, distribute and service one mixer will be

(a) ₹37.50
(b) ₹40.00
(c) ₹45.00
(d) ₹48.60

(Dec 2017)

Question: 60

Which of the following would take place if a company is able to reduce its variable cost?

Contribution Margin	Break-Even Point
(a) Increase	Increase
(b) Decrease	Decrease
(c) Increase	Decrease
(d) Decrease	Increase

(June 2017)

Question: 61

The following details relate to Product P-1 of a manufacturing company:

Level of activity (units)	1000	2000
Cost per unit (₹):		
Direct materials	4.00	4.00
Direct labour	3.00	3.00
Production Overheads	3.50	2.50
Selling Overheads	1.00	0.50
Total	11.50	10.00

The total fixed cost and variable cost per unit are:

	Total Fixed Cost (₹)	Variable Cost per unit (₹)
(a)	2,000	7.00

(b)	2,000	8.50
(c)	3,000	7.00
(d)	3,000	8.50

(June 2017)

Question: 62

A company makes a single product which it sells at ₹10 per unit. Fixed costs are ₹48,000 per month and the product has a contribution to sales ratio of 40%. In a period when actual sales were ₹1,40,000, the company's margin of safety in units was:

(a) 2000
(b) 3000
(c) 3500
(d) 4000

(June 2017)

Question: 63

A company has the capacity of production of 80000 units and presently it sells 20000 units at ₹100 each. The demand is sensitive to selling price and it has been observed that every reduction of ₹10 in selling price the demand is doubled. What should be the target cost at full capacity if profit margin on sales is taken at 25%?

(a) ₹58 lakhs
(b) ₹52 lakhs
(c) ₹48 lakhs
(d) ₹50 lakhs

(June 2017)

Question: 64

The following are cost data for two alternative ways of processing the clerical work for legal cases brought before the district court:

	Semi-automatic	Fully automatic
Monthly fixed costs (₹):		
Occupancy	15,000	15,000
Maintenance contract	5,000	10,000
Equipment lease	25,000	1,00,000
Unit variable cost (per report) (₹)	45000	125000
Supplies	80	20

Labour	60	20

The cost indifference point will be:

(a) 800 cases

(b) 850 cases

(c) 750 cases

(d) 700 cases

(June 2017)

1.	(B) 30%
2.	(B) 31,250 units
3.	(B) ₹ 12 lakhs
4.	(C) Contribution will be increased by ₹ 10 per hour for each extra hour of skilled labour that can be obtained.
5.	(C) ₹ 1,050
6.	(C) 30%
7.	(A) ₹ 900
8.	(A) ₹ 5
9.	(C) ₹ 4
10.	(D) ₹ 3,500
11.	(B) ₹ 75,000
12.	(C) ₹ 3,000
13.	(B) ₹ 440
14.	(A) ₹ 1,20,000
15.	(B) Growth
16.	(D) All of the above
17.	(A) ₹ 32,400
18.	(D) ₹ 40,000

19.	(B) ₹ 5
20.	(C) ₹ 82.50
21.	(B) ₹ 85 lakhs
22.	(B) ₹ 3,600
23.	(B) ₹ 13.10
24.	(B) ₹ 18,000
25.	(A) Market Driven Prices
26.	(A) 16.67%
27.	(B) ₹ 225 lakh
28.	(C) ₹ 6,000
29.	(D) ₹ 88,000
30.	(D) 55,000 units
31.	(D) desired ROI.
32.	(C) variable costing.
33.	(A) incremental revenues exceed incremental costs.
34.	(A) Target costing
35.	(C) more operating income will be reported under absorption costing than variable costing.
36.	(B) Target cost
37.	(A) maturity stage
38.	(D) Rs. 1,30,000
39.	(B) Opportunity cost
40.	(C) 30% **Explanation:** BEP = FC / P/V ratio = FC / BEP = Rs. 48,000 / 1,60,000 = **30%**

41. (D) zero

Explanation:

Opportunity cost is not an out of pocket cost.

It is the benefit given up by not selecting the next best alternative.

Therefore, answers A, B and C are incorrect and D is correct.

42. (C) ₹ 1050

Explanation:

ROI at 15% of total investment ₹ 15 lakhs

= ₹ 15,00,000 × 0.15

= **₹ 2,25,000**

Profit per unit of future output

= ₹ 2,25,000 / 500

= **₹ 450 per unit**

Therefore, target cost per unit

= Selling Price – Profit per unit

= ₹ 1,500 – ₹ 450

= **₹ 1,050 per unit**

43. (A) ₹ 4,20,000

Explanation: **Required Sales** = $\frac{\text{(FC + Desired Profit) / (1 – tax rate)}}{\text{Contribution / Sales}}$

$= \frac{₹50,000+90,000}{⅓}$

=₹4,20,000

44. (B) E, F, H & G

Explanation: Ranking of products would be in order of contribution per limiting factor, in relative value.

Particulars	**E**	**F**	**G**	**H**
SP (₹)	100	150	200	300

VC (₹)	50	70	100	125
Contribution per unit	50	80	100	175
RM / unit (kg)	1	2	5	7
Contribution per kg of RM (₹)	50	40	20	25
Rank	1	2	4	3

∴ **Correct order of ranking:** E, F, H & G

45. (C) Contribution will be increased by ₹10 per hour for each extra hour of skilled labour that can be obtained.
Explanation: A shadow price for a scarce resource is its opportunity cost.
It is the amount of contribution that would be lost if one unit less of that resource were available.
It is similarly the amount of additional contribution that would be earned if one unit more of that resource were available.
(This is on the assumption that the scarce resource is available at its normal variable cost.)

46. (B) Increase by 9.375%

Explanation:

S – V = C = Rs. 3,20,000 – 2,00,000 = **Rs. 1,20,000**

C/S ratio = (1,20,000 ÷ 3,20,000) × 100 = **37.5%**

New VC = Rs. 1,70,000

∴ C = Rs. 1,50,000

C/S ratio = (1,50,000 ÷ 3,20,000) × 100 = **46.875%**

∴ % increase in C = 46.875 – 37.5% = **9.375%**

47. (D) Rs.60
Explanation: as per following:

Demand	**Price (Rs.)**
20,000	100

	40,000 90 80,000 80 ∴ Target cost = Rs. 80 – (25% of 80) = **Rs. 60 /–**
48.	(A) Rs. 850 Explanation: Opportunity cost is the cost of next best alternative foregone. Between X and Y, **Y** has a better contribution i.e. Rs. 850 as against X (Rs. 1,500 – 700) = **Rs. 800**.
49.	(D) All of the above costs Explanation: Because all of the costs mentioned can be identified with specific goods/product and would be deducted from the selling price to determine the direct product profit.
50.	(B) Rs. 125 Explanation: Variable cost + (Fixed cost + Profit Desired) ÷ Volume = Rs. 100 + (1,20,000 + 30,000) ÷ 6000 = **Rs. 125 /–**
51.	(A) ₹ 10,50,000 **Explanation:** ₹ 10,50,000 = (85,00,000 × 30%) – 15,00,000 = **10,50,000** Or Sales × Contribution margin ratio or P.V. Ratio – Fixed Cost
52.	(B) 8,000 units **Explanation:** Contribution for 2000 units = 20,000 (difference in profits for two output levels) Hence, contribution per unit = **10** Substituting in equation: 1,00,000 = F + 10,000 Or F = **80,000** BEP = 80,000 ÷ 10 = **8000**
53.	(D) 350

	Explanation: Contribution = 1,06,000 + 74,000 = **1,80,000** Contribution / Unit = 1,80,000 ÷ 1200 = **150** Variable cost / Unit = 150 ÷ ¾ = **200** Selling price = **350**
54.	(B) ₹ 30,00,000 **Explanation:** Opportunity cost represents the next best alternative foregone. If B is chosen, only A is being foregone and hence the NPV of **30,00,000** is the present value of the opportunity lost.
55.	(A) 32193 **Explanation:** 1 double room = 1.3 single in terms of revenue. Capacity = 100 + 1.3 × 20 = 100 + 26 = **126 equivalent single rooms** Total Room Occupancy p.a. = 126 × 365 × 70% = **32193 days** Note: This can be arrived by other ways also, taking for example 70% of only single rooms and then double rooms, etc.
56.	(B) 40,000 units **Explanation:**SP 80 – VC 32 = **Contribution 48** F.C. **11,52,000** B.E.P. = 11,52,000 ÷ 48 = **24,000 units** MOS = **40%**; B.E.P. = **60%** ∴ Total sales = (24,000 × 100) ÷ 60 = **40,000 units**
57.	(C) ₹ 30,00,000

Explanation: Actual Sales – M.O.S. = BEP Sales

Sales = **₹ 50,00,000**

Less: Margin of safety 40% on sales = **₹ 20,00,000**

Break even sales = **₹ 30,00,000**

58. (B) to decrease by 1%

Explanation:

Particulars	Before installing new assets	After installing new assets
PBIT	₹ 19.20 lakhs	₹ 19.20 lakhs + ₹ 8.40 lakhs – ₹ 4.80 lakhs = **₹ 22.80 lakhs**
Value of Assets	₹ 96.00 lakhs	₹ 96.00 lakhs + ₹ 24.00 lakhs = **₹ 120.00 lakhs**
ROT	**20%**	**19%**

Conclusion: There will be a decrease of **1% in ROI** under the proposed dispensation.

59. (C) ₹ 45.00

Explanation: Projected sales (40,000 mixers × ₹ 60 per mixer) (A) = **₹ 24,00,000**

Less desired profit (15% of ₹ 40,00,000) (B) = **₹ 6,00,000**

Target Cost for 40,000 mixers (A – B) = **₹ 18,00,000**

Target cost per mixer (₹ 18,00,000 ÷ 40,000 mixer) = **₹ 45.00 per unit**

60. (C) Increase Decrease

Explanation:
Contribution margin = Sales Less Variable Cost

So, reduction in variable cost will increase contribution.

	BEP = FC / Contribution Margin Hence, increase in contribution will reduce BEP
61.	(D) 3,000 8.50 **Explanation:** Variable Cost per unit = 4.00 + 3.00 = **₹ 7.00** Total FC (included in Production Overheads and Selling Overheads) is as follows: **Units** — **Total OH** 1,000 — 4.50 × 1,000 = 4,500 2,000 — 3.00 × 2,000 = 6,000 Difference in Overhead = **₹ 1,500** Difference in Volume = **1,000** ∴ Variable per unit = **₹ 1.50** Add this to Variable cost per unit of ₹ 7.00. The Total variable cost = ₹ 1.50 + ₹ 7.00 = **₹ 8.50** Fixed Cost = 4,500 – (1,000 × 1.50) = 4,500 – 1,500 = **₹ 3,000**
62.	(A) 2000 **Explanation:** BEP = FC / C/S Ratio = 48,000 / 0.4 = **₹ 1,20,000 or 12,000 units** When sales are ₹ 1,40,000, the volume is ₹ 1,40,000 ÷ 10 = **₹ 14,000 units** ∴ Margin of Safety = 14,000 – 12,000 = **2,000 units**
63.	(C) ₹ 48 lakhs

Explanation:

Particulars	Details
Maximum Capacity	80,000 Units
Present Sale	20,000 Units @ ₹ 100 per Unit

Selling Price / Unit	Demand
100	20,000
90	40,000
80	80,000

Target Price = **₹ 80**

Target Cost / Unit
= 80 – 25% of Sales
= 80 – 20
= **₹ 60 per unit**

Total Target Cost
= 80,000 Units × ₹ 60 per unit
= **₹ 48 lakhs**

64. (A) 800 cases

Explanation:
Cost Indifference Point is calculated as follows:

Difference in monthly FC/Difference in unit VC
= (₹1,25,000–₹45,000)/(₹140–₹40) = ₹80,000/₹100 = 800 Cases

Chapter 4: Activity Based Management and Just in Time (JIT)

Question: 1

TONZ Ltd. manufactures two products BX and BZ using common material handling facility. The total budgeted material handling cost is ₹ 80,000. The other details are:

	Product BX	Product BZ
Number of units produced	20	40
Material moves per product line	15	5
Direct Labour hour per unit	100	100

Under Activity Based Costing System the material handling cost to be allocated to product BZ (per unit) would be

(A) ₹ 400

(B) ₹ 500

(C) ₹ 1,500

(D) None of the above

(Dec 2025)

Question: 2

Kanban Japanese System under JIT Approach ensures

(A) continuous supply of inventory or product.

(B) minimum and maximum level of stock to be maintained.

(C) inventory valuation.

(D) All of the above

(Dec 2025)

Question: 3

The companies that would benefit from backflush costing include companies

(A) which have fast manufacturing lead time.

(B) whose inventory vary from period to period.

(C) Company requires that audit trails.

(D) None of the above

(Jun 2025)

Question: 4

AON Ltd., a manufacturer of doors using a Throughput Costing System, is experiencing a bottleneck in its plant. Set up time at one of its Work Stations has been identified as the offender. The Manager (Finance) has proposed a plan to reduce the set up time at a cost of ₹ 5,24,000. The change will result in 500 additional doors. The Direct Labour costs are ₹ 2,400 per door and the cost of Direct Materials

is ₹ 8,000 per door. All units produced can be sold. If the change will result in an increase in the Throughput Contribution of ₹ 56,50,000, what will be the selling price per door?

(A) ₹ 19,300

(B) ₹ 18,500

(C) ₹ 13,700

(D) ₹ 21,700

(Jun 2025)

Question: 5

DASN Ltd., operates a throughput accounting system. The details of product MB per unit are as under:

	₹
Selling Price	50
Material Cost	20
Conversion Cost	15

If time on bottleneck resource is 10 minutes, what will be the return per hour for product MB?

(A) ₹ 300

(B) ₹ 270

(C) ₹ 180

(D) ₹ 150

(Dec 2024)

Question: 6

At KL Company, cost of personnel department has always been charged to production department based upon number of employees. Recently,opinion gathered from the department managers indicate that number of new hires might be better predictor of personnel cost. Total personnel department cost are ₹ 2,00,000.

Details of new hires and number of employees:

Department	A	B	C
Number of employees	30	270	100
Number of new hires	8	12	5

If number of new hires is considered the cost driver, what amount of cost will be allocated to Department A?

(A) ₹ 15,000

(B) ₹ 64,000

(C) ₹ 72,000

(D) ₹ 40,000

(Dec 2024)

Question: 7

Production overheads of XYZ Ltd. for 500 units of product X are:

Machine oriented activity cost: ₹ 1,35,400

Material ordering overheads: ₹ 69,570

Machine hours are 1·50 hours per unit and No. of material orders are 6 per unit.

What is the Machine Oriented Cost per unit and Material Ordering Cost per unit respectively?

(A) ₹ 270·80 & ₹ 139·14

(B) ₹ 180·53 & ₹ 23·19

(C) ₹ 23·19 & ₹ 180·53

(D) ₹ 139·14 & ₹ 270·80

(Jun 2024)

Question: 8

A factory of ROGO Ltd. has a key resource (bottleneck) of a facility M which is available for 48000 minutes per week. The time taken per unit of product A in facility M is 6 minutes. Last week's actual output was 5000 units of product A and actual factory cost (excluding material cost) was ₹ 1,20,000. What is throughput cost per week?

(A) ₹ 1,20,000

(B) ₹ 1,00,000

(C) ₹ 75,000

(D) Insufficient information

(Dec 2023)

Question: 9

The following tasks are associated with Activity Based Costing System:

(1) Allocation of Costs of Products

(2) Identification of Cost Pools

(3) Identification of Cost drivers

(4) Calculation of Pool rates The proper order of the preceeding tasks is

(A) (3), (2), (4), (1)

(B) (1), (3), (4), (2)

(C) (1), (2), (3), (4)

(D) (4), (3), (1), (2)

(Dec 2023)

Question: 10

RRS, a manufacturer of large windows, is experiencing a bottleneck in its plant. Setup time at one of its workstations has been identified as the culprit. A manager has proposed a plan to reduce setup time at a cost of ₹ 7,20,000. The change will result in 800 additional windows. The selling price per window is ₹ 18,000, direct labour costs are ₹ 3,000 per window, and the cost of direct materials is ₹ 7,000 per window. Assume all units produced can be sold. The change will result in an increase in the throughput contribution of _____.

(A) ₹ 64,00,000

(B) ₹ 88,00,000

(C) ₹ 56,80,000

(D) ₹ 1,44,00,000

(Dec 2023)

Question: 11

DYAM LTD. operates Throughput Accounting System. The details of product-D per unit are as under:

Selling Price	₹ 60
Material Cost	₹ 24
Conversion Cost	₹ 18
Time on Bottleneck resources	10 minutes

The return per hour for the product-D will be-

(A) ₹ 180

(B) ₹ 210

(C) ₹ 216

(D) ₹ 300

(June 2023)

Question: 12

OABIS LTD. operates an Activity Based Costing (ABC) system to attribute its overhead costs to cost objects. At this company, the cost of the personnel department has always been charged to production departments based upon number of employees. Recently, opinions gathered from the departmental heads indicate that the number of new hires might be a better predictor of personnel costs. Total personnel department costs are ₹ 3,00,000.

Particulars	Department		
	MP	BP	ZD
Number of Employees	20	200	80
Number of New hires	10	15	5

If number of new hires is considered the cost driver, the amount of personnel costs to be allocated to department BP to be

(A) ₹ 2,00,000

(B) ₹ 1,50,000

(C) ₹ 1,00,000

(D) None of the above

(June 2023)

Question: 13

Producing more non-bottleneck output

(A) creates more inventory and increases throughput contribution.

(B) creates more inventory but does not increase throughput contribution.

(C) creates less pressure for the bottleneck workstations.

(D) allows for the maximisation of overall contribution.

(June 2023)

Question: 14

At ROXIN LTD., cost of the personnel department has always been charged to the production department based upon number of employees. Recently opinion gathered from the department managers indicate that number of new hires might be better predictor of personnel cost. Total personnel department costs are ₹ 3,00,000.

Department	C	D	F
Number of Employees	40	300	160
Number of new Hires	10	32	8

If the number of new hire is considered the cost driver, what amount of cost will be allocated to Department D?

(A) ₹ 2,00,000

(B) ₹ 1,92,000

(C) ₹ 1,50,000

(D) ₹ 1,30,000

(June 2023)

Question: 15

AMRIT Ltd., a manufacturer of doors using a Throughput costing system, is experiencing a bottleneck in its plant.

Set-up time at one of its workstations has been identified as the offender.

The Manager (Finance) has proposed a plan to reduce the set-up time at a cost of ₹ 5,24,000.

The change will result in 500 additional doors.

The selling price per door is ₹ 19,300, Direct Labour costs are ₹ 2,400 per door and the cost of Direct Materials is ₹ 8,000 per door.

All units produced can be sold. The change will result in an increase in the Throughput Contribution of _____

(A) ₹ 60,20,000

(B) ₹ 56,50,000

(C) ₹ 44,50,000

(D) None of the above

19300
- 8000
11300 × 500

(Dec 2022)

Question: 16

Companies that would benefit from back-flush costing include companies –

(A) None of these.

(B) whose inventories vary from period to period.

(C) which have fast manufacturing lead times.

(D) companies that require audit trails.

(Dec 2021)

Question: 17

Which of the following is TRUE about the theory of constraints?

(A) TOC recognizes that lower inventories means slower response to customers.

(B) TOC recognizes that lowering inventory decreases carrying costs and thus decreases operating expenses and improves net income.

(C) TOC recognizes that lower inventories means more defects.

(D) TOC recognizes that EOQ is important.

(Dec 2021)

Question: 18

The operational activity of setting up equipment is classified as a –

(A) unit-level activity.

(B) facility-level activity.

(C) batch-level activity.

(D) product-level activity.

(Dec 2021)

Question: 19

A factory has a key resource (bottleneck) of Facility A which is available for 31,300 minutes per week. The time taken by per unit of Product X and Y in Facility A are 5 minutes and 10 minutes respectively. Last week's actual output was 4,750 units of Product X and 650 units of Product Y.

Actual factory cost was ₹78,250.

The throughput cost for the week would be:

(A) ₹75,625

(B) ₹76,225

(C) ₹77,875

(D) ₹79,375

(Dec 2019)

Question: 20

Which of the following is *not* a correct match?

Activity	Cost Drivers
(A) Production scheduling	Number of production runs
(B) Despatching	No. of Despatch orders
(C) Goods receiving	Goods received order
(D) Inspection	Machine hours

(Dec 2019)

Question: 21

XYZ Ltd. has the following alternative planned activity levels.

Level	E	F	G
Total cost	₹ 1,00,000	₹ 1,50,000	₹ 2,00,000
No. of units produced	5,000	10,000	15,000

If fixed overhead remains constant, then fixed overhead cost per unit at Level E is:

(A) ₹ 20

(B) ₹ 15

(C) ₹ 13.33

(D) ₹ 10

(June 2019)

Question: 22

A company manufactures and sells packaging machines. It recently introduced activity-based costing to refine its existing system. Each packaging machine requires direct materials costs of ₹ 50,000; 50 equipment parts; 12 machine hours; 15 assembly line hours and 4 inspection hours. The details about the cost pools, allocation bases and allocation rates are given below:

Indirect cost pool	Cost allocation base	Budgeted allocation rate
Material handling	No. of component parts	₹ 8 per part
Machining	Machine hours	₹ 68 per machine hour
Assembly	Assembly line hours	₹ 75 per assembly hour
Inspection	Inspection hours	₹ 104 per inspection hour

The company has received an order for 40 can-packaging machines from a customer. Using activity-based costing, indirect costs allocated to the order of the customer would be:

(A) ₹ 1,30,850

(B) ₹ 1,25,280

(C) ₹ 1,15,050

(D) ₹ 1,10,280

(June 2019)

Question: 23

Twin Ltd. uses JIT and back flush accounting. It does not use a raw material stock control account. During September 2018, 10,000 units were produced and sold. The standard cost per unit is Rs. 150 which includes materials of Rs. 60. During September 2018, Rs. 9,90,000 of conversion costs were incurred. The debit balance in cost of goods sold account for September 2018 was

(A) Rs. 14,00,000

(B) Rs. 14,80,000

(C) Rs. 15,90,000

(D) Rs. 16,20,000

(Dec 2018)

Question: 24

A company operates a standard absorption costing system. The budgeted fixed production overheads for the company for last year were Rs. 3,30,000 and budgeted output was 2,20,000 units. At the end of the company's financial year, the total of the fixed production overheads debited to the Fixed Production Overhead Control Account was Rs. 2,60,000 and the actual output achieved was 2,00,000 units. The under/over absorption of overhead was

(A) Rs. 40,000 over absorbed.

(B) Rs. 40,000 under absorbed.

(C) Rs. 50,000 over absorbed.

(D) Rs. 50,000 under absorbed.

(Dec 2018)

Question: 25

The following information relate to ABC

Activity level	Variable costs (₹)	Fixed costs (₹)
60%	12,000	20,000
80%	16,000	22,000

The differential cost for 20% capacity is

(a) ₹ 4,000

(b) ₹ 2,000

(c) ₹ 6,000

(d) ₹ 5,000

(June 2018)

Question: 26

A company operates an activity based costing (ABC) system to attribute its overhead costs to cost objects. In its budget for the year-ending 31st August 2018, the company expected to place a total of 2000 purchase orders at a total cost of ₹ 1,00,000. This activity and its related costs were budgeted to occur at a constant rate throughout the budget year which is divided into 13 four-week periods. During the four week period ended 30th Jun 2017, a total of 200 purchase orders were placed at a cost of ₹ 9,000. The over recovery of these costs for the four week period was

(a) ₹ 2,000

(b) ₹ 3,000

(c) ₹ 1,500

(d) ₹ 1,000

(June 2018)

Question: 27

A company uses traditional standard costing system. The inspection and set-up costs are actually ₹ 1,760 against a budget of ₹ 2,000. ABC system is being implemented and accordingly the number of batches is identified as the cost driver for inspection and set up. The budgeted production is 10,000 units in batches of 1,000 units whereas actually 9,000 units were produced in 11 batches. The cost per batch under ABC system will be

(a) ₹ 160
(b) ₹ 200
(c) ₹ 180
(d) ₹ 220

(Dec 2017)

Question: 28

When you wait until the manufacture of a product has been completed and then record all of the related issuances of inventory from stock that were required to create the product, it is called

(a) Forensic Accounting
(b) Back-flush Accounting
(c) Tax Accounting
(d) Lean Accounting

(Dec 2017)

Question: 29

The following tasks are associated with ABC system:

I. Allocation of costs to products
II. Identification of cost pools
III. Identification of cost drivers
IV. Calculation of pool rates

The proper order of the preceding tasks is:

(a) III, II, IV, I
(b) I, II, III, IV
(c) III, IV, II, I
(d) IV, III, II, I

(June 2017)

Question: 30

Backflush costing is most likely to be used when:

(a) Management desires sequential tracking of costs
(b) A Just-in-Time inventory philosophy has been adopted
(c) The company carries significant amount of inventory
(d) Actual production costs are debited to work-in-progress

(June 2017)

1.	(B) ₹ 500

2.	(A) continuous supply of inventory or product.
3.	(A) which have fast manufacturing lead time.
4.	(A) ₹ 19,300
5.	(C) ₹ 180
6.	(B) ₹ 64,000
7.	(A) ₹ 270.80 & ₹ 139.14
8.	(C) ₹ 75,000
9.	(A) (3), (2), (4), (1)
10.	(B) ₹ 88,00,000
11.	(C) ₹ 216
12.	(B) ₹ 1,50,000
13.	(B) creates more inventory but does not increase throughput contribution
14.	(B) ₹ 1,92,000
15.	(B) ₹ 56,50,000
16.	(C) which have fast manufacturing lead times.
17.	(B) TOC recognizes that lowering inventory decreases carrying costs and thus decreases operating expenses and improves net income.
18.	(C) batch-level activity.
19.	(A) ₹ 75,625 **Explanation:** Cost per Factory Minute = Total Factory Cost / Minutes Available = ₹ 78,250 / 31,300 = **₹ 2.50** Standard Minutes of throughput for the week = (4750 × 5) + (650 × 10) = **30,250 minutes**

	Therefore, throughput cost for the week = 30,250 × ₹ 2.50 = **₹ 75,625**
20.	(D) Inspection Machine Hours **Explanation:** Inspection hours, and not machine hours, drive the cost of inspection.
21.	(D) ₹ 10 Explanation: Change in Costs (B – A) = ₹ 50,000 Change in Units (B – A) = ₹ 5,000 VC per unit = ₹ 50,000 ÷ 5,000 = **₹ 10** Total Cost at A = ₹ 1,00,000 VC : 5,000 × ₹ 10 = ₹ 50,000 Total FC = ₹ 50,000 ÷ 5,000 units = **₹ 10 per unit**
22.	(D) ₹ 1,10,280 Explanation: **Indirect costs per machine:**

Activity	Calculation	Amount (₹)
Material handling	₹ 8 × 50	400
Machining	₹ 68 × 12	816
Assembly	₹ 75 × 15	1,125
Inspection	₹ 104 × 4	416
Total		**2,757**

For the order:
₹ 2,757 × 40 = **₹ 1,10,280**

23. (C) Rs.15,90,000

Explanation:

Particulars	Rs.
Standard cost of goods sold (10,000 units @ Rs. 150)	15,00,000
Less: Std. material cost (10,000 @ Rs. 60)	6,00,000
	9,00,000
Standard conversion cost	
Conversion cost incurred	9,90,000
Excess charged to cost of goods sold a/c (debit)	90,000

∴ Total debit balance of cost of goods sold Account
= Rs. 15,00,000 + 90,000
= **Rs. 15,90,000**

24. (A) Rs. 40,000 over absorbed.

Explanation:

Overhead Absorption Rate = Rs. 3,30,000 ÷ 2,20,000 units = **Rs. 1.50 / unit**

Overhead Absorbed = 2,00,000 @ Rs. 1.50 = **Rs. 3,00,000**

Actual overhead = **Rs. 2,60,000**

Over absorbed overhead = **Rs. 40,000**

25. (C) ₹ 6,000
Explanation:
Differential Costs = Differences in Fixed and Variable Cost
= 4000 + 2,000 = **6,000**

26. (D) ₹ 1,000

Explanation:
For 2,000 purchase orders, cost budgeted is 1 lac.

	For 200, corresponding amount would be 10,000. But actual = 9,000. Hence over recovered is 10,000 – 9000 = **1000** Or Cost driver rate for order = 1,00,000 ÷ 2000 = **50 per order** Cost recovered = 50 × 200 = **10,000** Actual = **9,000** Over recovery = **1000**
27.	(B) ₹ 200 Explanation:Number of batches under ABC = 9000 ÷ 1000 = 9 Std. Cost under ABC = Budg Cost / Batch × ABC number of batches = ₹ 200 × 9 = ₹ 1800

Particulars	Details
Production	9000 Units
Number of batches	9
Cost / Batch	₹ 200

28.	(B) Back-flush Accounting
29.	(A) III, II, IV, I **Explanation:** Because cost is allocated based on the cost pool rates. So, whole process starts with identification of cost drivers followed by identification of cost pools, determination of rates and then allocation.
30.	(B) A Just-in-Time inventory philosophy has been adopted **Explanation:** A Just-in-Time inventory philosophy has been adopted.

The reason for this is that JIT assumes zero inventory for raw materials, work-in-progress and finished goods and the system of backflush accounting records the transaction only at the termination of the production and sales cycle.

Chapter 5: Evaluating Performance

Question: 1

YEN Ltd., adopting Standard Costing System in its factory, budgets for fixed overhead of ₹ 50,000 and production of 10,000 units for a month of year 2025. Actual production is 9,500 units. If the fixed overhead cost is increased to ₹ 52,000, the fixed overhead volume variance will be

(A) ₹ 1,000 (Adv.)

(B) ₹ 2,000 (Fav.)

(C) ₹ 2,500 (Fav.)

(D) ₹ 2,500 (Adv.)

(Dec 2025)

Question: 2

Standard Cost and Budgeted Cost are

(A) interrelated but not interdependent.

(B) interdependent but not interrelated.

(C) interrelated and interdependent.

(D) None of the above

(Dec 2025)

Question: 3

In a factory where standard costing system is followed, the production department consumed 1100 kgs of material @ ₹ 8 per kg for product X resulting in material price variance of ₹ 2,200 (Fav) and material usage variance of ₹ 1,000 (Adv). What is the standard material cost of actual production of product X?

(A) ₹ 11,000

(B) ₹ 20,000

(C) ₹ 14,000

(D) ₹ 10,000

(Jun 2025)

Question: 4

In a factory where standard costing system is followed the production department consumed 1100 kgs of a material @ ₹ 8 per kg for product X resulting in material price variance of ₹ 2200(Fav) and material usage variance of ₹ 1000(Adv).What is the standard material cost of actual production of product X?

(A) ₹ 11,000

(B) ₹ 20,000

(C) ₹ 14,000

(D) ₹ 10,000

(Dec 2024)

Question: 5

BONAS Ltd. adopting a Standard Costing System provides the following information pertaining to Direct Materials for the month of April, 2024.

Particulars	₹
Standard price per unit	₹ 7·20
Actual quantity purchased in units	3200
Standard quantity allowed for actual production in units	2900
Material Price Variance on purchase (FAV)	₹ 960

What will be the actual purchase price per unit?

(A) ₹ 8·50

(B) ₹ 7·80

(C) ₹ 6·90

(D) ₹ 6·50

(Jun 2024)

Question: 6

SHIND Ltd., a manufacturing company using Standard Costing system, has the following information pertaining to a normal production of 15000 units of a product for the month of April. Standard factory overheads rates are based on a normal volume of one standard direct labour hour per unit for the month of April.

Standard factory overheads rate per Direct Labour hour are :

Variable	₹ 15
Fixed	₹ 8
	₹ 23

Units actually produced during the month 14000 units

Actual factory overhead cost incurred (including ₹ 87,000 Fixed) ; ₹ 2,90,000

Actual direct labour hours 14000 hours

What will be the Variable Overhead Expenditure variance?

(A) ₹ 9,000 (Adv.)

(B) ₹ 8,000 (Fav.)

(C) ₹ 8,000 (Adv.)

(D) ₹ 7,000 (Fav.)

(Dec 2023)

Question: 7

A Co. producing output X and Y uses standard costing. The standard overhead contents of each product is:

X: ₹ 3 per unit and Y: ₹ 2.25 per unit.

The budgeted overhead is ₹ 860 and budgeted time is 3440 hours.

Actual Output:
X 200 units and Y 100 units.
Actual time: 3200 hours
Actual overhead: ₹ 875. Compute Overhead Volume variance.
(A) ₹ 35 (A)
(B) ₹ 35 (F)
(C) ₹ 25 (F)
(D) ₹ 15 (A)

(Dec 2023)

Question: 8

In a factory of BIVU LTD., where standard costing system is followed, the production department allowed 1,200 hours of standard labour hours @ ₹ 16 per labour hour for the Actual Production of 1,000 units of Product-AMU resulting in Labour rate variance of ₹ 15,000 (FAV). The Actual Labour rate per hour was ₹ 14.50. How many actual labour hours were worked during the month for BIVU LTD.?
(A) 20,000 hours
(B) 12,000 hours
(C) 10,000 hours
(D) 9,000 hours

(June 2023)

Question: 9

BOSAN LTD. using standard costing system provides the following information pertaining to Direct Labour for its product JUM for the month of May, 2023:

Standard Direct Labour Rate per hour:	₹ 16
Actual Direct Labour Rate per hour	₹ 14.50
Labour Rate variance	₹ 15,000 (Fav)
Standard hours allowed for Actual production	8000 hours

Actual output is 800 units.
How many Direct Labour hours were worked during the month of May, 2023?
(A) 12,000 hours
(B) 11,000 hours
(C) 10,000 hours
(D) None of the above

(June 2023)

Question: 10

RON Ltd., a manufacturer of product CEMO using a standard costing system provides the following information pertaining to the Direct Materials for the month of November 2022:

Particulars	Details
1 tonne of material input yields standard output of (units)	1,00,000

Standard price of material per kg (₹)	20
The Actual quantity of material used (Tonnes)	10
Actual price of material per kg (₹)	21
Actual output obtained for the month (units)	9,00,000

Material cost variance will be —

(A) ₹ 40,000 (Adv)

(B) ₹ 40,000 (Fav)

(C) ₹ 30,000 (Adv)

(D) ₹ 25,000 (Adv)

(Dec 2022)

Question: 11

A manufacturing company has the following information pertaining to a normal monthly production of 10,000 units of a product.

Standard factory overhead rates are based on a normal monthly volume of one standard direct hour per unit.

Standard factory overhead rates per direct labor hour are:	
Fixed	Rs. 6.00
Variable	Rs. 10.00
Units actually produced in current month	9,000 units
Actual factory overhead costs incurred (Includes Rs. 70,000 fixed)	Rs. 1,56,000
Actual direct labor hours	9,000 hours

The variable overhead spending variance is –

(A) Rs. 0

(B) Rs. 10,000 (F)

(C) Rs. 4,000 (F)

(D) Rs. 86,000 (A)

SR AH SH
10 9000 9000

(Dec 2021)

Question: 12

The higher the actual hours worked,

(A) The lower the capacity usage ratio.

(B) The higher the capacity usage ratio.

(C) The lower the capacity utilization ratio.

(D) The higher the capacity utilization ratio.

(CSR-ARH)AR

SRAH - ARAH

90000 - 86000 = 4(FAVE)

and vice versa.

4. **Capacity Usage Ratio:** Capacity Usage Ratio is the relationship between the budgeted number of working hours and the maximum possible number of working hours in a budget period.

Capacity Usage Ratio	=	(Budgeted Hours ÷ Maximum Possible Hours in Budget) × 100

80000 100000hr

Capacity usage ratio indicates the extent to which the budgeted hours have actually been utilised.

1. **Capacity Utilisation Ratio:** It is the relationship between actual hours in a budget period and the budgeted working hours in the period.

Capacity Utilisation Ratio	=	(Actual Hours ÷ Budgeted Hours) × 100

ec 2019)

Question: 13

Max Ltd. fixes the inter-divisional transfer prices for its products on the basis of cost plus a return on investment in the division.

The budget for division X for 2019–20 appears as under:

Particulars	₹
Fixed assets	5,00,000
Current assets	3,00,000
Debtors	2,00,000
Annual fixed cost of the division	8,00,000
Variable cost per unit of the product	10
Budgeted volume	4,00,000 units per year
Desired ROI	28%

Transfer price for division X is:

(A) ₹12.70

(B) ₹10.70

(C) ₹8.70

(D) ₹14.70

(Dec 2019)

Question: 14

A manufacturing company uses two types of materials, X and Y, for manufacture of a standard product.

The following information is given:

	Standard Mix		Actual Mix	
Materials X	120 Kg @ ₹5 =	₹600	112 Kg @ ₹5 =	₹560
Materials Y	80 Kg @ ₹10 =	₹800	88 Kg @ ₹10 =	₹880
Total	200	₹1,400	200	₹1440
Loss	60 (30%)		50(25%)	
Output	140	₹1,400	150	₹1440

Direct Materials Mix Variance is:

(A) ₹40 (fav.)

(B) ₹40 (unfav.)

(C) ₹80 (fav.)

(D) ₹80 (unfav.)

(Dec 2019)

Question: 15

AB Ltd. uses standard cost system. The following information pertains to direct labour for Product X for the month of March, 2019:

Particulars	Amount
Standard rate per hour	₹ 8
Actual rate per hour	₹ 8.40
Standard hours allowed for actual production	2,000 hours
Labour Efficiency variance	₹ 1,600 (Adverse)

What were the actual hours worked?

(A) 1,800

(B) 1,810

(C) 2,200

(D) 2,190

(June 2019)

Question: 16

Uniform Costing may not be successfully applied in the following case:

(A) In a single enterprise having a number of branches, each of which manufactures the same set of products with the same facilities.

(B) In a number of entities in the same industry bound by a trade association.

(C) In a number of units across different geographical locations manufacturing one or more of a given set of products.

(D) In different branches of the same company, each branch making a different product using a unique process.

(June 2019)

Question: 17

In a factory where standard costing system is followed, the production department consumed 1100 kgs of a material @ ₹ 8 per kg for product X resulting in material price variance of ₹ 2200 (Fav) and material usage variance of ₹ 1000 (Adv). What is the standard material cost of actual production of product X?

(a) 11,000

(b) 20,000

(c) 14,000

(d) 10,000

(June 2018)

Question: 18

The following figures are extracted from the books of a company:

Budgeted O/H ₹10,000 (Fixed ₹6,000, Variable ₹4,000)

Budgeted Hours 2000

Actual O/H ₹10,400 (Fixed ₹6,100, Variable ₹4,300)

Actual Hours 2100

Variable O/H cost variance and Fixed O/H cost variance will be:

(a) 100 (A) and 200 (A)

(b) 100 (F) and 200 (F)

(c) 100 (A) and 200 (F)

(d) 200 (A) and 100 (F)

(Dec 2017)

Question: 19

The information relating to the direct material cost of a company is as follows:

Standard price per unit	₹7.20
Actual quantity purchased in units	1600
Standard quantity allowed for actual production in units	1450
Material price variance on purchase (Favourable)	₹480

What is the actual purchase price per unit?

(a) ₹7.50

(b) ₹6.40

(c) ₹6.50

(d) ₹6.90

(June 2017)

Question: 20

The preparation and use of standard cost, their comparison with actual costs and the measurement and analysis of variances to originating causes is defined as:

(a) Marginal Costing

(b) Standard Costing

(c) Throughput Costing

(d) Kaizen Costing

(June 2017)

1.	(D) ₹ 2,500 (Adv.)
2.	(A) interrelated but not interdependent.
3.	(D) ₹ 10,000

4. (D) ₹ 10,000

5. (C) ₹ 6.90

6. (D) ₹ 7,000 (Fav.)

7. (A) ₹ 35 (A)

8. (C) 10,000 hours

9. (C) 10,000 hours

10. (C) ₹ 30,000 (Adv)

11. (C) Rs. 4,000 (F)

12. (D) The higher the capacity utilization ratio.

Explanation:
Capacity utilization ratio = **Actual Hours / Budgeted Hours**

So, the capacity utilization ratio would be higher.

13. (A) ₹ 12.70

Explanation:

Particulars	**Per unit (₹)**
VC	10
FC (₹ 8,00,000 ÷ 4,00,000)	2
Investment : (FA + CA + Debtors) = ₹ 10,00,000	
Return = ₹ 10,00,000 × 0.28 ÷ 4,00,000	0.70

	TP for Div. X | **12.70**
14.	(B) ₹ 40 (unfav.)
15.	(C) 2,200 Explanation: Labour Efficiency Variance = (ST – AT) × SR or, (2,000 – AT) × ₹ 8 = (–) ₹ 1,600 or, AT = 7,600 ÷ 8 = **2,200 hours**
16.	(D) In different branches of the same company, each branch making a different product using a unique process. Explanation: Though the entity is the same, different products using different (unique) process cannot follow uniform costing.
17.	(D) 10,000 **Explanation:** Actual Cost + Favourable Cost Variance = Standard Cost 1100 × 8 + 2200 – 1000 = 8800 + 1200 = **10,000**
18.	(C) 100 (A) and 200 (F) **Explanation:**Variable O/H Cost variance = Recovered O/H – Actual O/H = 4200 – 4300 = **100 (A)** Fixed O/H Cost variance = 6300 – 6100 = **200 (F).**
19.	(D) ₹6.90 **Explanation:** Material Price Variance (MPV) = Standard cost of Actual Quantity – Actual Cost

	480 = 7.20 × 1,600 – Actual Cost or, Actual Cost = 11,520 – 480 = **11,040** Actual Price / Unit = 11,040 ÷ 1,600 = **₹ 6.90**
20.	(B) Standard Costing **Explanation:** Because standard costing only involves the process described.

Chapter 6: Linear Programming

Question: 1

If (50, 0), (50, 14), (25, 50) and (13, 50) are the Corner Points of the feasible region of the Linear Programming problem, Maximize $Z = 2x_1 + 3x_2$, then maximum value of the objective function will be

(A) ₹ 200

(B) ₹ 270

(C) ₹ 240

(D) ₹ 120

(Dec 2025)

Question: 2

The constraints in a linear programming Model are —

(A) Included in the Objective function

(B) Costs

(C) Scarce Resources

(D) Dependent Variables

(Dec 2022)

Question: 3

Which of the following is a valid constraint for a linear programming problem?

(A) $3x^2 + 4x + 1 = 0$

(B) $5x_1 + 2x_2 \leq 10$

(C) $4x_1 + 3x_2 > 7$

(D) $(12x_1 + 4x_2) / 3x_2 \leq 8x_1$

(June 2019)

1.	(A) ₹ 200
2.	(C) Scarce Resources
3.	(B) $5x_1 + 2x_2 \leq 10$ Explanation: Other options do not conform to linearity or fundamental of constraints.

Chapter 7: Transportation Problem

Question: 1

The highest negative opportunity cost value in an unused cell of a Transportation Matrix is chosen to improve the current solution because –

(A) It represents maximum possible cost reduction per unit.

(B) It ensures no violation of Rim Condition.

(C) It represents per unit cost improvement.

(D) Either one of the above

(Dec 2023)

Question: 2

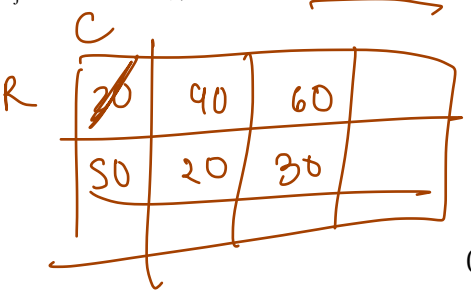
In a transportation matrix (where R_i are rows and C_j are columns), the second allocation under the North West Corner Rule can be

(A) R1C2

(B) R1C3

(C) R2C3

(D) None of the above



(June 2023)

Question: 3

In a transportation matrix (where Ri are rows and Cj are columns), the second allocation under the North West Corner Rule can be –

(A) R1C2

(B) None of these

(C) R2C3

(D) R1C3

(Dec 2021)

1.	(A) It represents maximum possible cost reduction per unit
2.	(A) R1C2
3.	(A) R1C2

Chapter 8: Assignment

Question: 1

In marking assignments which of the following should be preferred?

(A) Only row having single zero

(B) Only column having single zero.

(C) Column having more than one zero

(D) Only row/column having single zero

(Jun 2025)

Question: 2

When a maximisation assignment problem is converted to minimisation problem, the resultant matrix is called:

(A) Profit Matrix.

(B) Regret Matrix.

(C) Cost Matrix.

(D) Dummy Matrix.

(Jun 2024)

Question: 3

Choose the most appropriate answer to the following questions giving justification. 2x2=4 1 mark will be awarded for correct answer and 1 mark for justification.

(i) Assignment Problem can be considered as a particular case of

(A) Transportation problem

(B) Sequencing problem

(C) Queuing problem

(D) All of the above

The Assignment Problem is a special case of the Transportation Problem because:

Each source (person/machine) has a supply of 1.
Each destination (job/task) has a demand of 1.
Every person is assigned to exactly one job, and every job is assigned to exactly one person.

(June 2023)

1.	(D) Only row/column having single zero
2.	(B) Regret Matrix
3.	(A) Transportation problem

Chapter 9: Game Theory

Question: 1

The pay-off matrix of a game for the Company S and Company T is given below:

	Company T			
Company S	25	12	10	30
	15	13	6	8
	30	5	8	6

Using Maximin-Minimax principle, what will be the value of the game?

(A) 15

(B) 12

(C) 10

(D) None of the above

(Dec 2025)

Question: 2

The pay-off matrix of game for Firm A and Firm B is given below:

	FIRM B			
FIRM A	30	15	12	35
	20	13	8	10
	40	4	11	6

What is the value of game (using Maximin - Minimax Principle)?

(A) 6

(B) 10

(C) 12

(D) None of the above

(Jun 2025)

Question: 3

The Pay-off Matrix of a game is given below:

	Player B		
Player A	3	-1	4
	-1	8	2
	16	8	6

(Dec 2024)

What is the value of Game to Player A (using Maximin–Minimax Principle)?

(A) 6

(B) 8

(C) 16

(D) None of the above

(Jun 2024)

Question: 4

When fewer than m+n–1 shipments exist in a feasible solution

(A) the problem is capacitated.

(B) an occupied cell must be created artificially.

(C) a dummy source must be created.

(D) the problem is unbalanced.

(Dec 2023)

Question: 5

The value of the game of

		Player B	
		B1	B2
Player A	A1	4	6
	A2	-10	10

is _____.

Fill in the above.

(A) 4

(B) 6

(C) 8

(D) None of the above

(Dec 2023)

1.	(C) 10
2.	(C) 12
3.	(A) 6
4.	(B) an occupied cell must be created artificially
5.	(A) 4

Chapter 10: Simulation

Question: 1

Children's Park in a city with various signals and crossing is the simulated model of city traffic.This is a type of

(A) Analogue Simulation

(B) Computer Simulation

(C) Monte Carlo Simulation

(D) Both (A) and (C) of the above

(Dec 2024)

Question: 2

The important step(s) required for simulation approach in solving a problem is/are:

(A) Test and validate the Model

(B) Design the experiment

(C) Conduct the experiment

(D) All of the above

(Jun 2024)

Question: 3

The drive-up window of a fast-food operation was being studied using simulation for a variety of operating characteristics. As part of the study, data was collected on Order Processing Time as given in the following table. Using the first two digits of the Random Numbers, determine the processing time that would be used to simulate the fifth sample.

Processing time (Minutes)	1	2	3	4
Probability	0.30	0.45	0.20	0.05

Customer	1	2	3	4	5	6	7	8	9
Random No	1048	2236	2413	4216	3757	1501	4657	4836	9309

(A) 2 minutes

(B) 4 minutes

(C) 1 minute

(D) 3 minutes

(Dec 2023)

1.	(A) Analogue Simulation
2.	(D) All of the above

3.	(A) 2 minutes

Chapter 11: Network Analysis

Question: 1

A PERT Network has four activities on its Critical Path. The Standard Deviation of each activity on the Critical path is 4 days. What will be the standard deviation of the CRITICAL PATH?

(A) 8 Days

(B) 4 days

(C) 2 days

(D) None of the above

(Jun 2025)

Question: 2

Optimistic and pessimistic time of an activity respectively are 4 days and 16 days. Variance of the duration of the activity will be:—

(A) 4 days

(B) 2 days

(C) 3 days

(D) None of the above

(Dec 2024)

Question: 3

The normal duration and normal cost of an activity of project BOZ are 12 days and ₹ 400 respectively. The cost slope is ₹ 80 per day. if the crash cost of the activity is ₹ 560, what will be the crash duration of the activity?

(A) 12 Days

(B) 10 Days

(C) 8 Days

(D) 6 Days

(Dec 2024)

Question: 4

The expected time for an activity of Project LM is 20 days. If the most likely and pessimistic time are 19 days and 28 days, what will be the variance of the activity?

(A) 1·36

(B) 1·78

(C) 4

(D) 0

(Jun 2024)

Question: 5

The expected time for an activity of Project A is 10 days. If the optimistic and pessimistic times are 8 days and 12 days respectively, the most likely time estimate will be:

(A) 2 days

(B) 5 days

(C) 10 days

(D) None of the above

(Dec 2023)

Question: 6

The slack time of tail event of activity Z of a project is 2 days. If the Total Float and Free float of the activity Z are 10 days and 7 days respectively, the Independent float of activity Z will be:

(A) 5 days

(B) 6 days

(C) 7 days

(D) 2 days

(Dec 2023)

Question: 7

The expected time for an activity of BS Project (t_e) is 19 days. If the Optimistic(t_o) and the most likely (t_m) times are 15 days and 18 days the pessimistic (t_p)time estimate will be

(A) 17 days

(B) 21 days

(C) 27 days

(D) 28 days

(June 2023)

Question: 8

In PERT CHART, the Activity time distribution is

(A) Normal

(B) Binomial

(C) Beta

(D) Poisson

(June 2023)

Question: 9

A PERT network has only two activities on its critical path. The Standard Deviation of these activities are 6 and 8 respectively. What is the Standard Deviation of the Project completion time?

(A) 7

(B) 10

(C) 14

(D) 100

(Dec 2022)

Question: 10

The following will be the appropriate action to finish a project early –

(A) Crash activities on the non-critical path so that they become critical.

(B) Crash activities on the non-critical paths so that they remain non-critical.

(C) Crash activities on the critical path so that they become non-critical.

(D) Crash activities on the critical paths such that the critical paths remain critical.

(Dec 2021)

Question: 11

In a PERT network, the optimistic time for a particular activity is 9 weeks and the pessimistic time is 21 weeks.

Which one of the following is the best estimate of the standard deviation for the activity?

(A) 12

(B) 9

(C) 6

(D) 2

(Dec 2019)

Question: 12

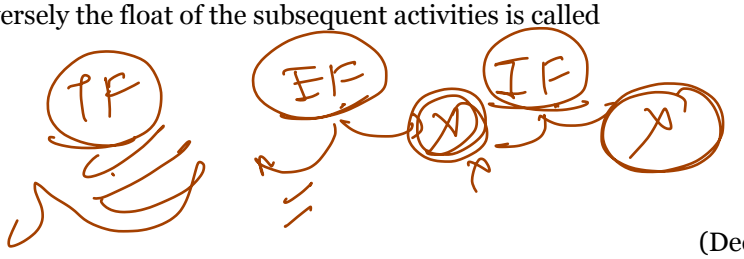
In the context of Critical Path Analysis, the portion of the float of an activity which cannot be consumed without affecting adversely the float of the subsequent activities is called

(A) Free float

(B) Interfering float

(C) Independent float

(D) Total float



(Dec 2018)

Question: 13

In CPA (Critical Path Analysis) which of the following is not a correct step in sequence?

(A) Understanding the logic of the system under consideration

(B) Constructing the network

(C) Providing estimates for activity duration.

(D) Implementing and controlling the network.

(Dec 2018)

Question: 14

Which of the following is correct in the context of network analysis?

(a) There can be one or more activities without a predecessor in a network.

(b) Where two activities have the same start and end events, the end event of one activity is numbered differently and then connected by a dummy to the original start event.

(c) When crashing is carried out, the non-critical paths have to remain non critical.

(d) If the critical path is longer than the other paths, the project may be completed by using a path having a shorter duration.

(June 2018)

1.	(A) 8 days
2.	(A) 4 days
3.	(B) 10 Days
4.	(C) 4

5.	(C) 10 days
6.	(A) 5 days
7.	(C) 27 days
8.	(C) Beta
9.	(B) 10
10.	(D) Crash activities on the critical paths such that the critical paths remain critical.
11.	(D) 2 **Explanation:** Standard Deviation equals (pessimistic time minus optimistic time) / 6 that is = (21 – 9) / 6 = **2**
12.	(B) Interfering float Explanation: Interfering float is that part of the total float which causes a reduction in the float of the successor activities. It is the difference between the latest finish time of the activity in question and the earliest starting time of the following activity or zero, whichever is larger.
13.	(D) Implementing and controlling the net work. Explanation: Because step no. 4 i.e. (d) should be satisfying the objectives. Implementing and controlling the network would be the final step.
14.	(A) There can be one or more activities without a predecessor in a network. **Explanation:** More than 1 activity can begin at the first node, say 1 – 2, 1 – 3, 1 – 4, etc. Each of these will have no predecessor.

Chapter 12: Learning Curve

Question: 1

An operation of KON Ltd. has a Learning Curve 90% and the first unit produced took 30 minutes. The labour cost is ₹ 120 per hour. How much should the second unit labour cost?

(A) None of the below options

(B) ₹ 60

(C) ₹ 54

(D) ₹ 48

(Dec 2025)

Question: 2

When 24 hours is required to produce a condenser of a particular type then the time required to produce the 16th unit with 85% Learning Curve is

(A) between 9 and 10 hours.

(B) between 12 and 14 hours.

(C) between 15 and 17 hours.

(D) between 18 and 20 hours.

(Jun 2025)

Question: 3

LONAX Ltd., a manufacturing company has developed a new product and just completed the manufacture of first 16 units of the product. If the first unit took 6 hours to manufacture and the first 16 units together took 62.9856 hours to produce, the Learning Curve (LC) rate would be

(A) 80 %

(B) 85 %

(C) 90 %

(D) 95 %

(Dec 2024)

Question: 4

DOXT Ltd., a manufacturing company, is preparing a quotation for a new product-D. The time taken for the first unit is 30 hours. The company expects 85% learning curve (Index is - 0·2345). The company desires that the quotation should be based on the time taken for the final output within the Learning period which is expected to end after the company has produced 200 units. What will be the time per unit of product to be used for the quotation?

[Given: $199^{(-0.2345)} = 0.28901$, $200^{(-0.2345)} = 0.28867$ and $201^{(-0.2345)} = 0.28834$]

(A) 6·63 hours

(B) 10·34 hours

(C) 11·50 hours

(D) None of the above

(Jun 2024)

Question: 5

MOB Ltd., has developed a new product and just completed the manufacture of first four units of the product. If the first unit took 4 hours to manufacture and the first 4 units together took 11.56 hours to produce, the Learning Curve (L.C) rate would be:

(A) 69.50 %
(B) 75.00 %
(C) 80.10 %
(D) 85.00 %

(Dec 2023)

Question: 6

An employee of ROB Ltd. took 200 minutes to complete the first set up on a new machine. Using a 90% incremental unit time learning model indicates that the second set up on the new machine is expected to take-

(A) 160 minutes
(B) 120 minutes
(C) 100 minutes
(D) 80 minutes

(Dec 2023)

Question: 7

SOX LTD. manufacturers a product-Z whose time for the first unit is 80 hours. If it experiences a learning curve of 80% what will be the time to be taken to complete the 8th unit of a 12 unit batch involved in the Assembly line?

(A) 43.50 hours
(B) 42.71 hours
(C) 40.96 hours
(D) 38.50 hours

(June 2023)

Question: 8

SIXT Ltd. has developed a new product and just completed the manufacture of the first four units of the product. The first unit took 3 hours to manufacture and the first four units together took 8.3667 hours to produce. The learning curve rate is

(A) 83.50%
(B) 75.00%
(C) 60.65%
(D) 58.50%

(Dec 2022)

Question: 9

A learning curve is a function –

(A) where unit costs increase as productivity increases.
(B) that increases at a greater rate as workers become more familiar with their tasks.
(C) that is linear.
(D) that measures the decline in labor-hours per unit due to workers becoming better at a job.

(Dec 2021)

Question: 10

To complete the first setup on a new machine took an employee 200 minutes. Using an 80% incremental unit-time learning model indicates that the second setup on the new machine is expected to take –

(A) 120 minutes.

(B) 160 minutes.

(C) 60 minutes.

(D) 80 minutes.

(Dec 2021)

Question: 11

X is a factory making a certain product where learning curve ratio of 80% and 90% apply respectively for two equally paid workers, A and B.

(A) The labour cost of manufacturing the 4th product will be more for A.

(B) The labour cost of manufacturing the 4th product will be more for B.

(C) The labour cost is the same for the fourth product.

(D) Nothing can be said about the specific product since learning applies ratio to the average quantity of the product.

(Dec 2019)

Question: 12

S Ltd. recently sold an order of 50 units having the following costs:

Particulars	₹
Direct materials	1,500
Direct labour (1,000 hours @ ₹8.50)	8,500
Variable overhead (1,000 hours @ ₹4.00)[1]	4,000
Fixed overhead[2]	1,400
Total	15,400

1. Allocated on the basis of direct labour-hours.
2. Allocated at the rate of 10% of variable cost.

The company has now been requested to prepare a bid for 150 units of the same product.

If an 80% learning curve is applicable, Stone Isle's total cost on this order would be:

(A) ₹ 38,500

(B) ₹ 37,950

(C) ₹ 26,400

(D) ₹ 31,790

(June 2019)

Question: 13

S Ltd. manufactures a product whose time for the first unit is 1000 hours. It experiences a learning curve of 80%. What will be the total time taken in hours for unit 5 to 8?

(A) 4096 hours
(B) 3200 hours
(C) 1536 hours
(D) 2000 hours

(Dec 2018)

Question: 14

An operation has a 90% learning curve and the first unit produced took 28 minutes. The labour cost is ₹ 20 per hour. How much should the second unit cost?

(a) ₹ 9.80
(b) ₹ 7.60
(c) ₹ 8.40
(d) ₹ 6.60

(June 2018)

Question: 15

For a Learning Curve percentage of 72%, the time to be taken to complete the 4th unit of a 12-unit job involved in the assembly line, if the initial unit requires 80 hours, will be

(a) 43.50 hrs
(b) 41.47 hrs
(c) 46.71 hrs
(d) 40.95 hrs

(Dec 2017)

1.	(D) ₹ 48
2.	(B) between 12 and 14 hours
3.	(C) 90 %
4.	(A) 6.63 hours
5.	(D) 85.00%
6.	(A) 160 minutes
7.	(C) 40.96 hours
8.	(A) 83.50%
9.	(D) that measures the decline in labor-hours per unit due to workers becoming better at a job.
10.	(A) 120 minutes.

11.	(B) The labour cost of manufacturing the 4th product will be more for B **Explanation:** The labour cost of manufacturing the 4th product will be more for B since B will take more time per unit of product.
12.	(C) ₹ 26,400 Explanation: Cumulative hours 200 × (20 × 0.8 × 0.8) = 2560 Less: 50 × 20 = 1000 Net hours for 150 units = **1560** Cost: Direct Materials = 150 × 30 = 4,500 Direct Labour = 1560 × 8.50 = 13,260 Variable Overhead = 1560 × 4 = 6,240 Total Variable Cost = **24,000** Allocated Fixed OH = 10% = 2,400 **Estimated Cost of the Order = 26,400**
13.	(C) 1536 hours Explanation: At **80% Learning Curve**, the total time for **8 units** will be 8 × 512 i.e. **4096 hours** and for **4 units** it is 4 × 640 i.e. **2560 hours** Hence the time taken for units **5 to 8** will be **1536 hours** (4096 – 2560)
14.	(B) ₹ 7.60 **Explanation:** 1st unit = 28 min. Average time p.u. for 2 units = 0.9 × 28 = **25.2** Total time for 2 units = 25.2 × 2 = **50.4** Time for second unit = 50.4 – 28 = **22.4 minutes** Cost for second unit = 22.4 × 20 ₹ / hr. / 60 minutes = **7.47** Since, (b) is close to 7.47, **b is acceptable**. Otherwise, none of the given data.
15.	(B) 41.47 hrs **Explanation:**At **72% Learning Curve**, T-4 = Time taken by the 4th Unit = 80 (.72)(.72) = **41.47 hrs.** **Note:** In the arithmetic method followed above, every time the number of repetitions doubles, the time to perform the activity is reduced by the Learning Curve Coefficient.

Chapter 13: Business Applications of Maxima and Minima

Question: 1

The Cost function of TNZ PIC is given by C = $X^3 - 4X^2 + 7X$ in terms of Output X. Assess at what level of output the average cost is minimum.

(A) 3
(B) 2
(C) 4
(D) None of the above

(Jun 2025)

Question: 2

The average cost function for a product is given by AC = $(1/3)x^2 + x + 8$, in terms of output x. If the marginal cost is ₹ 23 what will be the number of units (Output)?

(A) 3
(B) 4
(C) 5
(D) 0

(Dec 2024)

Question: 3

Total Revenue of ZOM Ltd., from the Sales of the quantity (units) is given by the equation R = $(1/3)x^3 - x^2 + 10x + 5$. If the Marginal revenue is 25, what will be number of quantity (in units)?

(A) 6
(B) 4
(C) 5
(D) None of the above

(Jun 2024)

1.	(B) 2
2.	(A) 3
3.	(C) 5

Chapter 14: Business Forecasting Models

Question: 1

In Simple Exponential Smoothing forecast to give higher weightage to recent demand information, the smoothing constant must be close to

(A) -1

(B) 0

(C) 0.3

(D) 1

(Dec 2024)

Question: 2

Which of the following is not a part of qualitative type of forecasting techniques?

(A) Survey Technique

(B) Barometric Technique

(C) Exponential Smoothing

(D) Delphi Technique

(Dec 2023)

1.	(D) 1
2.	(C) Exponential Smoothing

Chapter 15: Introduction to Tools of Data Analytics

Question: 1

Which of the following is / are feature(s) of Zoho Analytics Tools?

(A) BI and Analytics platform

(B) Deployable in the cloud as well as on premises server

(C) Both (A) and (B) of the above

(D) Rich Accounting Solution for small business

(Dec 2025)

Question: 2

Which of the following is related to Financial Data Analytics?

(A) Value driver analytics

(B) Financial ratio analytics

(C) Predictive sales analysis

(D) All of the above

(Jun 2025)

Question: 3

Python can be used on Linux, Windows, Macintosh, Solaris and many more. What category of language features of Python it comes under?

(A) Robust

(B) Embeddable

(C) High Level Language

(D) Platform Independent

(Jun 2025)

Question: 4

Which one of the following is not a Spread Sheet?

(A) Quip

(B) MS Excel

(C) E views

(D) Google Sheets

(Dec 2024)

Question: 5

Script Ends - is related to which type of Programming Language ?

(A) SPSS

(B) Python

(C) SAS

(D) R Programming

(Dec 2024)

Question: 6

Which of the following is/are not the features of Jedox tool?

(A) Deployable in the cloud as well as on premise server or Hybrid.
(B) BI and Analytics platform.
(C) Allows users to easily create and share powerful reports within minutes.
(D) (B) and (C) only

(Jun 2024)

Question: 7

Analysis of a dataset has revealed the fact that profit of a business has reduced for the financial year 2022-23. What category of data analytics it comes under?

(A) Descriptive Analytics
(B) Predictive Analytics
(C) Diagnostic Analytics
(D) Prescriptive Analytics

(Jun 2024)

Question: 8

Which of the following has no relation to Business Intelligence?

(A) A set of business analytics solutions to retrieve, analyse and transform data into useful business sights.
(B) Visualisation Tools are primarily BI Tools.
(C) ABS Glue is a tool used for the purpose of Business Intelligence.
(D) Embedded Analytics is an important part of any Business Intelligence tool.

(Dec 2023)

Question: 9

Which of the following is/are the Financial analytics tools?

(A) Hypper Anna
(B) Jedox
(C) Net Suite
(D) All of the above

(Dec 2023)

1.	(C) Both (A) and (B) of the above
2.	(D) All of the above
3.	(D) Platform Independent
4.	(C) E views
5.	(B) Python
6.	(D) (B) and (C) only

7.	(A) Descriptive Analytics
8.	(C) ABS Glue is a tool used for the purpose of Business Intelligence
9.	(D) All of the above